

KAUHAJOKI KUJA-KOKKO, Finland

WMO: 027690

Lat: 62.4133N

Lon: 22.1836E

Elev: 93

StdP: 100.21

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.5	(c) -21.2	(d) -26.9	(e) 0.3	(f) -24.4	(g) -23.5	(h) 0.5	(i) -21.0	(j) 10.4	(k) -0.4	(l) 9.4	(m) -0.3	(n) 0.9	(o) 160	(p) 0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.0	(c) 26.2	(d) 18.3	(e) 24.2	(f) 17.2	(g) 22.3	(h) 16.5	(i) 19.8	(j) 23.9	(k) 18.5	(l) 22.4	(m) 17.4	(n) 21.2	(o) 3.4	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.4	13.4	21.7	16.9	12.2	20.3	15.7	11.3	19.0	57.1	24.1	52.7	22.3	49.1	21.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 8.4	(b) 7.4	(c) 6.6	(e) -27.0	(f) 28.0	(g) 4.0	(h) 2.8	(i) -29.9	(j) 30.1	(k) -32.2	(l) 31.7	(m) -34.4	(n) 33.3	(o) -37.3	(p) 35.4		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-7.4	-5.9	-2.9	2.8	9.1	12.7	15.8	14.1	10.2	3.8	0.5	-3.3
	DBStd	9.04	6.64	6.70	5.11	3.50	4.36	3.41	3.24	3.12	3.35	3.74	4.41	5.92
	HDD10.0	2612	539	446	399	216	69	11	1	3	38	194	286	411
	HDD18.3	5187	797	679	658	465	288	172	92	136	245	451	536	669
	CDD10.0	492	0	0	0	1	41	93	182	131	43	2	0	0
	CDD18.3	24	0	0	0	0	1	3	14	6	0	0	0	0
	CDH23.3	262	0	0	0	0	22	45	156	38	0	0	0	0
	CDH26.7	40	0	0	0	0	1	4	29	6	0	0	0	0
(14) Wind	WSAvg	3.1	2.9	3.2	3.3	3.1	3.3	3.2	2.9	3.0	2.9	3.1	3.2	3.3
(15) Precipitation	PrecAvg	653	49	36	32	29	44	63	78	77	60	68	58	58
	PrecMax	821	104	74	57	56	80	137	141	136	135	153	127	105
	PrecMin	514	9	1	7	4	10	19	5	18	19	14	10	18
	PrecStd	80	22	19	11	14	21	25	32	31	34	32	24	26
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.5	5.7	9.2	19.1	25.9	27.1	29.7	27.4	20.6	13.7	10.2	6.9
		MCWB	3.7	3.4	4.3	10.2	15.3	17.8	20.2	20.5	16.2	11.6	9.4	6.0
	2%	DB	3.0	3.9	6.3	14.3	23.2	24.5	27.0	23.9	18.5	11.7	8.2	5.0
		MCWB	2.0	2.5	3.5	8.0	14.6	16.6	19.1	18.0	14.7	10.3	7.5	4.2
	5%	DB	1.7	2.5	4.6	11.7	20.8	22.2	24.9	21.9	17.0	10.7	6.8	3.7
		MCWB	1.1	1.7	2.4	6.7	13.3	15.5	18.1	16.6	13.7	9.5	6.2	3.0
	10%	DB	0.8	1.5	3.3	9.4	18.0	19.9	22.9	20.1	15.7	9.5	5.7	2.7
		MCWB	0.3	0.9	1.7	5.5	11.9	14.4	17.6	15.7	13.2	8.4	5.2	2.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	3.6	5.2	11.3	17.0	19.7	21.7	21.4	16.8	12.1	9.4	6.2
		MCDB	4.5	4.8	7.6	16.2	23.3	23.8	26.7	25.1	19.8	13.1	10.1	6.8
	2%	WB	2.3	2.6	3.7	8.9	15.7	18.0	20.3	19.1	15.4	10.9	7.6	4.4
		MCDB	2.8	3.7	5.7	14.3	21.4	23.1	24.5	22.3	17.4	11.5	8.2	5.0
	5%	WB	1.2	1.8	2.8	7.4	14.1	16.4	19.1	17.6	14.5	9.7	6.3	3.1
		MCDB	1.5	2.5	4.3	10.8	19.6	20.8	23.0	20.8	16.3	10.6	6.7	3.5
	10%	WB	0.4	1.0	1.9	5.9	12.2	15.1	18.1	16.5	13.6	8.5	5.3	2.2
		MCDB	0.8	1.4	3.2	9.2	17.6	19.2	22.1	19.2	15.3	9.3	5.7	2.6
(35) Mean Daily Temperature Range	MDBR		6.2	6.5	8.1	8.9	10.7	9.7	10.0	9.6	8.2	6.1	4.0	4.9
	5% DB	MCDBR	4.2	5.2	7.8	14.3	15.4	14.3	13.5	12.4	10.4	7.3	4.6	4.3
		MCWBR	3.9	4.5	5.4	8.6	7.6	7.5	6.5	6.8	6.9	6.0	4.3	4.2
	5% WB	MCDBR	4.0	5.3	6.8	12.3	13.6	11.7	11.4	9.7	8.7	6.5	4.5	4.4
		MCWBR	3.9	4.6	5.0	7.9	7.2	6.6	6.3	5.7	6.2	5.8	4.3	4.4
(40) Clear-Sky Solar Irradiance	taub		0.245	0.297	0.320	0.360	0.356	0.352	0.375	0.361	0.337	0.319	0.265	0.352
	taud		2.080	2.193	2.282	2.333	2.421	2.455	2.412	2.442	2.501	2.388	2.156	2.565
	Ebn at Noon		415	646	783	825	863	872	837	819	773	629	385	249
	Edh at Noon		34	65	86	101	100	99	101	90	71	54	31	20
(44) All-Sky Solar Radiation	RadAvg		0.18	0.74	2.25	3.73	5.12	5.56	5.19	3.98	2.36	0.94	0.24	0.09
	RadStd		0.03	0.16	0.25	0.44	0.52	0.43	0.48	0.39	0.26	0.19	0.04	0.01

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (42 neighbors)		+0.46	N/A	N/A	N/A	N/A	N/A	N/A	-130	N/A	N/A	N/A

Nomenclature: See separate page